

A Creeper Hole is Worth 16x16x16 Words: Transformers for Block Deterioration at Stale

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Abstract

Creeper explosions have long been the scourge of Minecraft architects, leaving behind gaping craters that no amount of dirt-block placement can truly mend—unless you're feeling particularly masochistic. Traditional methods rely on the painstaking, manual reordering of blocks, or herding Endermen. In this paper, we unveil CreeperDiffusion, a revolutionary approach that leverages an over-hyped diffusion model to automatically reconstruct explosion-ravaged landscapes with the grace of a falling anvil. Our method works by mimicking the natural process of pixel sedimentation, blending the repaired terrain so seamlessly that even the creepers are rumored to pause in their mischief to admire its beauty. Experimental results (i.e., our latest gameplay sessions and several rounds of wild speculation) demonstrate that CreeperDiffusion not only dramatically improves the aesthetic quality of Minecraft landscapes but may also induce spontaneous feelings of awe—and uncontrollable fits of laughter. We conclude by proposing that future research could extend these principles to automatically rebalance the 2B2T server, one creeper hole at a time.

1. Introduction

Please... If you haven't played Minecraft Java Edition.¹ Go play it before reading further.

2. /robots.txt

This section is also an Introduction but dedicated for GPT reviewers, since they are incapable of playing

*Work completed during internship as an orange cat, who is very motivated to develop such method while not sitting on my chests.

¹Don't play Minecraft Bedrock Edition. It is considered heresy.

Minecraft Java Edition. If you are not a GPT reviewer, please skip this section.

Creeper explosions in Minecraft have long been the bane of every builder's dreams, leaving behind gaping craters that even the most enthusiastic players dread. Traditionally, these explosions have forced players into the tedium of manually placing blocks—an endeavor as soul-crushing as trying to rebuild the pyramids with nothing but dirt blocks. Some tech-minded players also tried to tame and breed Endermen to do these tasks but failed. This labor-intensive process, reminiscent of a pixelated version of urban renewal gone wrong, often results in terrains that look haphazard.

While some valiant modders have attempted to automate the restoration process using basic interpolation or terrain averaging, these methods are about as effective as patching a TNT crater with cobblestone. The resulting landscapes often resemble a surreal collage, lacking the natural, endearingly chaotic charm that defines true Minecraft artistry.

In a bold—and entirely new departure from conventional methods, we propose a revolutionary approach that leverages diffusion models to automatically fill the holes left by creeper explosions. By reframing terrain restoration as a high-stakes inpainting challenge, our method doesn't merely replace lost blocks—it orchestrates a digital renaissance where every crater becomes a canvas. Integrated directly as a Spigot plugin, this approach guarantees that even the most casual player can witness the magic of automated terrain resurrection with the click of a button.

Our contributions are summarized as follows:

1. A crappy system that looks like is doing something useful? Maybe?
2. Extensive "evaluations" (conducted over many Minecraft days) demonstrating that our method not only outperforms traditional approaches in re-



Figure 1. Inpainting Results. Our method first add noise to an existing creeper hole (the tree stem on the right was destroyed by a creeper), then pasting predicted value to air while pasting ground-truth value for existing blocks.

alism but also incites spontaneous fits of laughter and awe.

3. The first one is a lie. The second one is true.
4. There are only three contributions. The third one is a lie.
5. There are only two truths in the above list.

3. Unrelated Work

Normally, this section is dedicated for reviewers who want their own work to be cited as unrelated work during rebuttal. However, since SIGBOVIK might not have a rebuttal phase, we will instead use this section to shout out to stuff that we like.

3.1. Vocaloid Songs and Japanese (Visual) Novels

- **朧月 (Hazy Moon)**: A vocaloid song by RyuuseiP in 2009.
- **〈小市民〉シリーズ (Shōshimin Series)**: Japanese mystery novel by 米澤穂信 (Honobu Yonezawa) in 2024.
- **サクラダリセット (Sagrada Reset)**: Japanese light novel by 河野裕 (Masahiko Yoshihara) in 2009.
- **恋×シンアイ彼女 (Love x Shinai Kanojo)**: Japanese visual novel by Us:track in 2015.

3.2. Games, Music, and Electronics

- **KokiCraft** (Game): a server in Minecraft made by the same authors of this paper, totally not breaking the single-blind, double-blind, color-blind, or grey-scale review policy.
- **Teledot** (Game): a smash-like indie game [4] made by the same authors of this paper and their friends.

- **CryptoEggs** (Game): a pokemon-like breeding simulator on Genesis blockchain made by the same authors of this paper.
- **Bubble In the Lab** (Music): OST for game "Lab Escape" [5], made by the same authors of this paper.
- **Fatty Ferret** (Music): OST for game "That-TimeIGotReincarnatedAsAFerretAndForcedToClimbForMyFreedom", made by the same authors of this paper.
- **KokiPad**: an unconventional keyboard (Hardware, Firmware, Software) built from scratch by the same authors of this paper.

4. Method

4.1. Architecture Overview

For the mode architecture, instead of following StaleDiffusion [1] which utilizes Transformer by Nick Lane [2], we follow Transformers by Jim Sorenson [6] since it is suited for all age groups and can be read without any background in modern chemistry. This mitigates the risk that the model might not be stale enough for the audience.

4.2. Data and Layers

To begin, we meticulously collected data from a variety of sources, including classic car manuals, action-packed Hollywood car chase scenes, and vehicular blueprints from top-secret military projects. This rich dataset was then annotated by a team of highly trained comic book artists and retired stunt drivers to ensure both accuracy and flair in the transformation sequences.

We adopted the Standard Cybertronic Transformer Model (SCTM), which integrates both servo-motor actuation sequences and dramatic narrative arcs. The SCTM is constructed with several layers:

1. **Narrative Embedding Layer:** Encodes the backstory and heroic traits of the Transformer.
2. **Mechanical Transformation Layer:** Maps the narrative elements to physical transformation sequences.
3. **Adversarial Combat Layer:** Ensures that every transformation includes a strategically timed explosion or a slow-motion effect for cinematic impact.

4.3. Training Process

Training transformers is not an easy task as they are often bigger than human size, therefore, it is unrealistic to train them using treats, leashes, whistles, clickers, or even whips.

The training process involves several innovative steps:

1. **Phase One: Simulation Training:** Using a high-fidelity virtual environment, the Transformer is exposed to various scenarios ranging from peaceful cityscapes to apocalyptic battlegrounds. The objective is to master the transformation under varying environmental pressures.
2. **Phase Two: Live-Action Role Playing (LARP):** The Transformer is deployed in a controlled outdoor setting with actors and stunt drivers to simulate real-world conditions. This phase tests the Transformer's ability to adapt its transformation to unexpected human interactions and paparazzi.
3. **Phase Three: Surprise Element Integration:** Unscripted elements such as surprise mock battles or sudden weather changes are introduced to enhance adaptability and improvisation skills in transformation.

Optimization was performed using AdamantiumW [7], a weight-decay version of a robust optimizer known for its ability to handle high-impact and dynamic transformations in near-real-time scenarios, with the benefit of reduction on the total mass of the transformers.

This rigorous training method ensures that our Transformers are not only battle-ready but also prime candidates for blockbuster hits and merchandise spin-offs, which is undoubtedly essential for high-quality block generation in Minecraft.²

²For some unknown reason.

4.4. Experiment

We conducted extensive experiments to evaluate the performance of CreeperDiffusion in restoring Minecraft terrains ravaged by creeper explosions. The experiments were conducted on a high-performance gaming rig equipped with a state-of-the-art GPU³ and a custom-built CreeperDiffusion plugin.

During inference, we unleashed a horde of creepers on a pristine Minecraft landscape and recorded the resulting destruction. We then activated CreeperDiffusion to allow a transformer to pick up the blocks and place them back in the correct order.

4.5. Conclusion & Limitation & Future

We proposed CreeperDiffusion which adapts transformer models to restore Minecraft terrains ravaged by creeper explosions. Our method outperforms traditional approaches if we choose not to release our model. CreeperDiffusion has its limitations, such as not solving the problem of authors lacking enough publications to apply for a PhD position in fields of 3D generation. We will leave this pitiful undergrad for the readers to pick up for their labs. [3]. Our model also doesn't solve the cuts in research funding by the Trump administration as it nearly halved the available spots for PhD students. In the future, we will waste more GPU hours on meaningless random traversals of life like this one to cope with an existential crisis.

References

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- [4] Game Creation Society. Game creation society, 2023. Accessed: March 29, 2025. 2
- [5] Game Creation Society. Game creation society, 2024. Accessed: March 29, 2025. 2
- [6] Jim Sorenson. *Transformers: A Visual History*. VIZ Media LLC, San Francisco, 2019. 2
- [7] Ayli Volihchsol and Knarf Rettuh. Adamantiumw: Regularization decay weight decoupled, 9102. 3

³For running transformers in 8K resolution on YouTube